

# Report - Building Materials

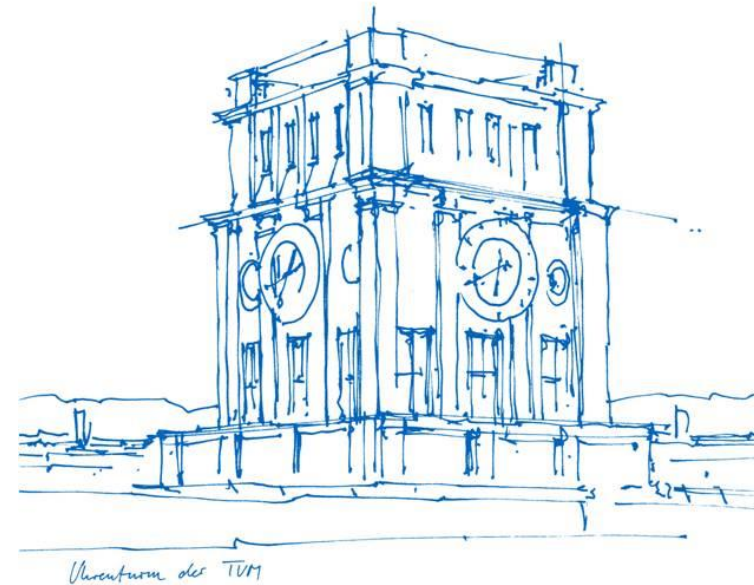
Felix Förster

Technische Universität München

TUM School of Computation, Information and Technology

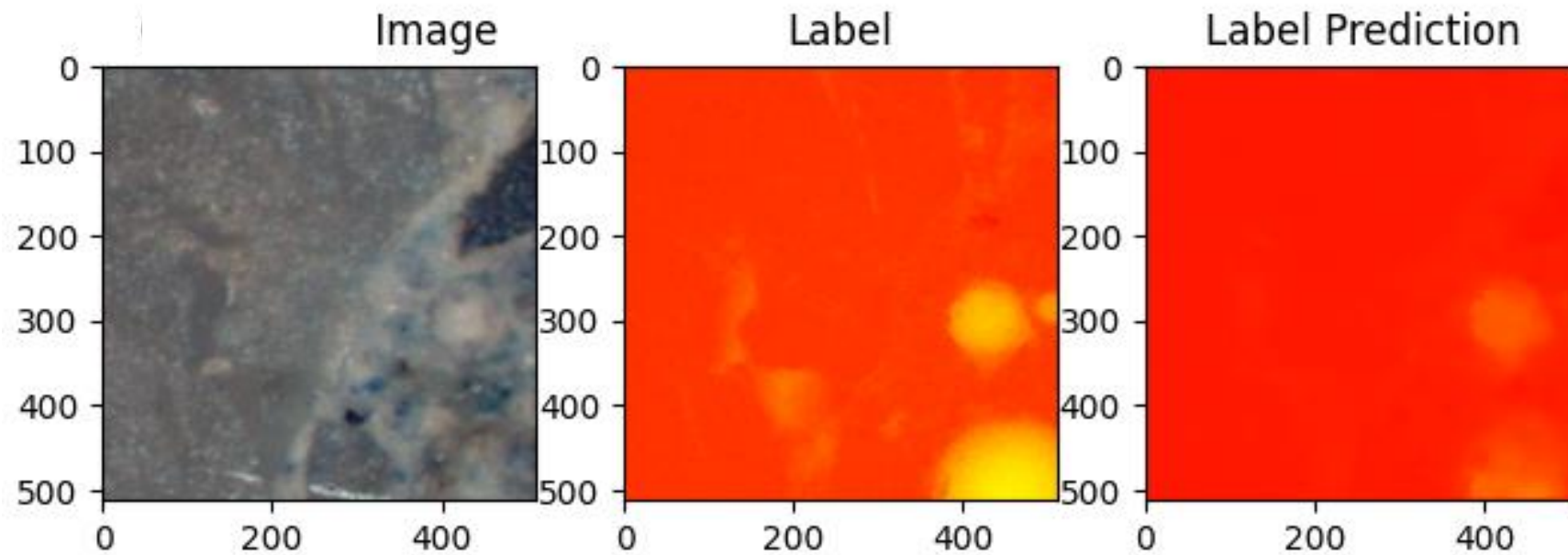
Lehrstuhl für Computer Vision and Artificial Intelligence

21. March 2024

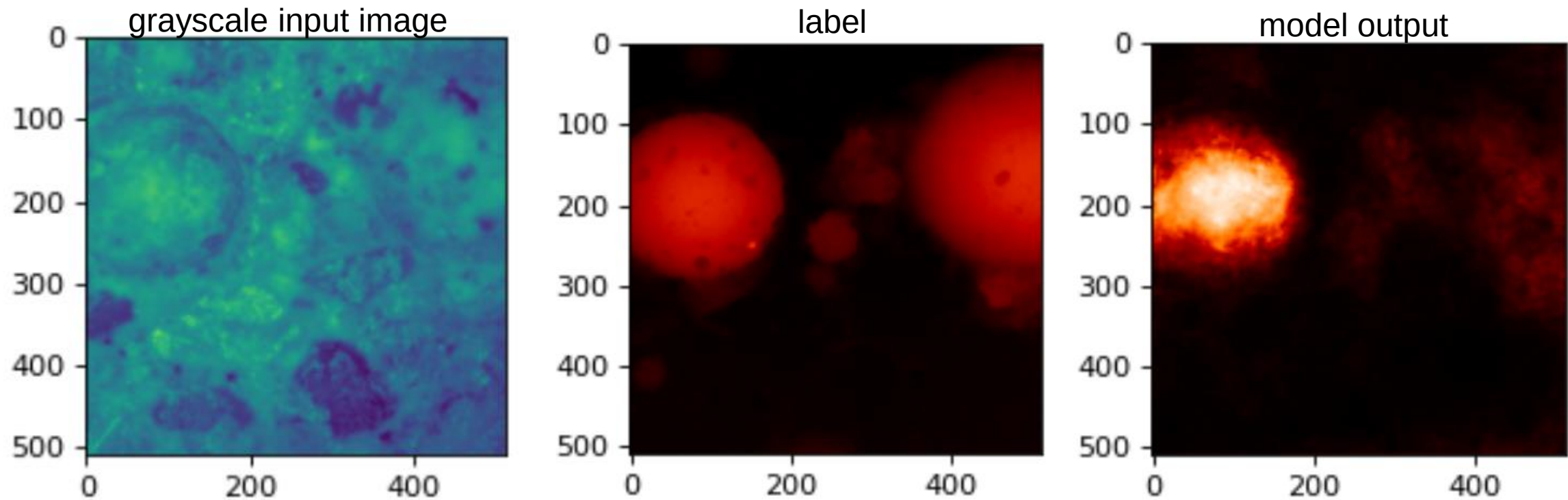


# Recap – Status 21. December 2023

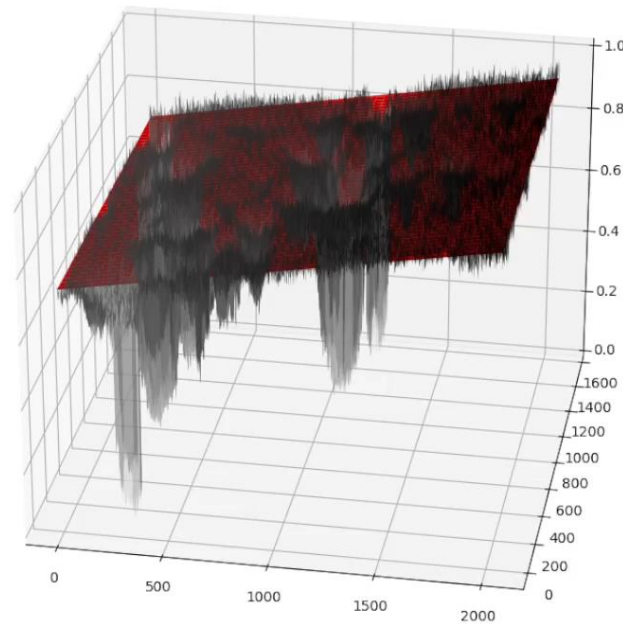
## Relative height: results



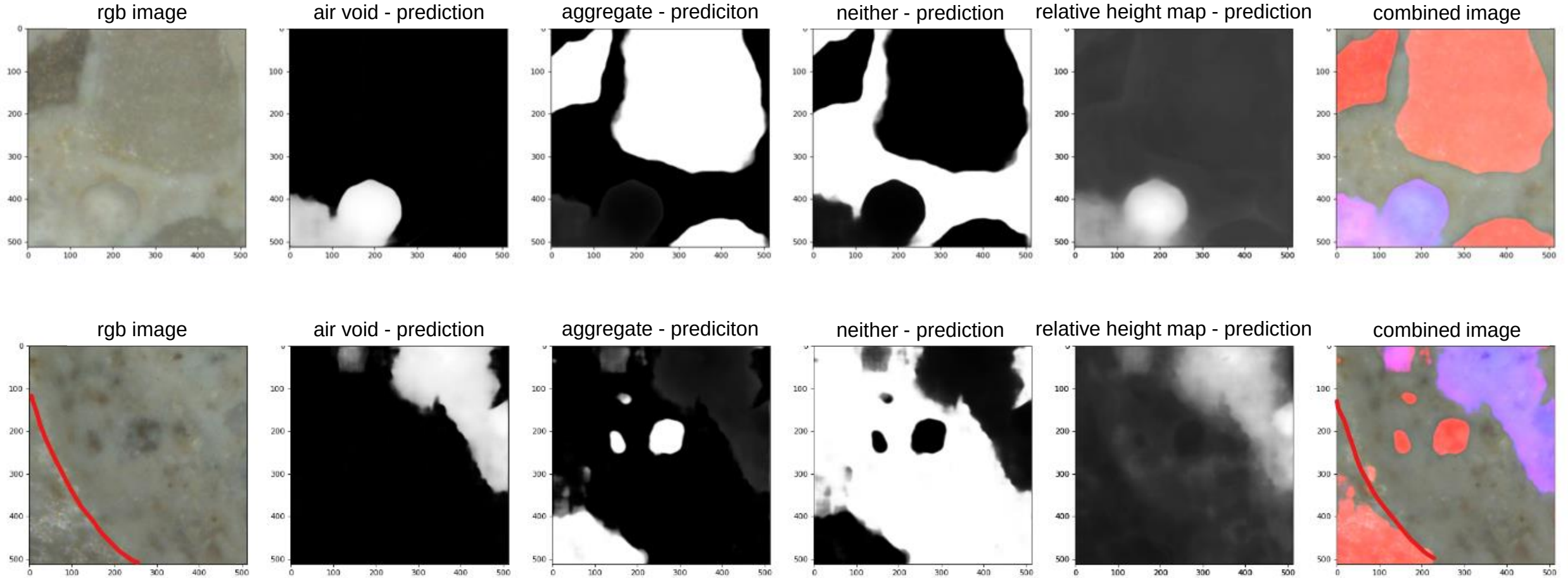
# Absolute height



# Compensating tilted planes – relative height



# Mask prediction



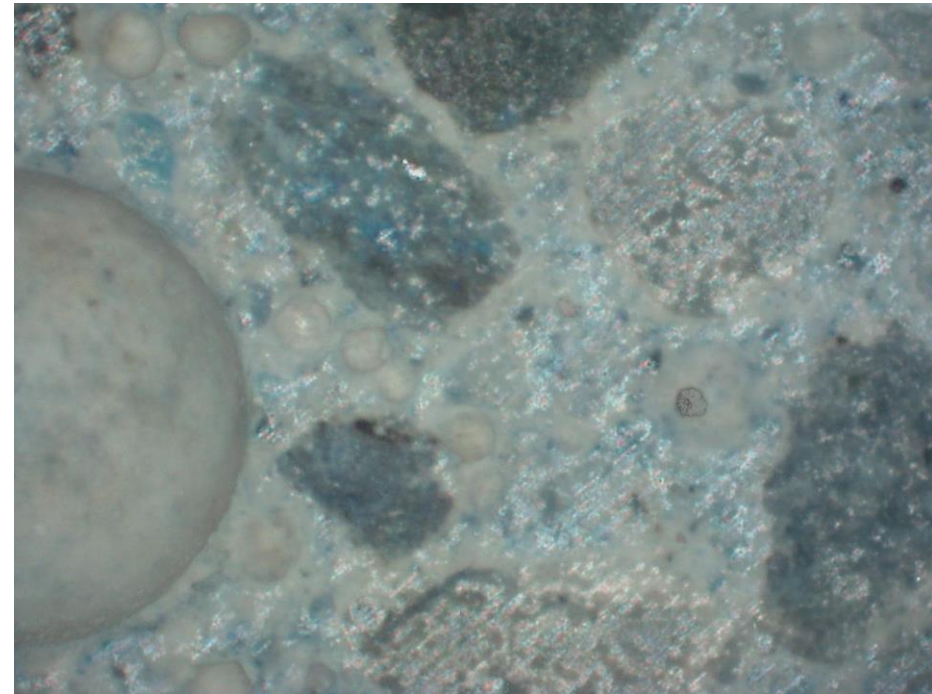


# Only aggregate classifier

only aggregate / without height



contains height

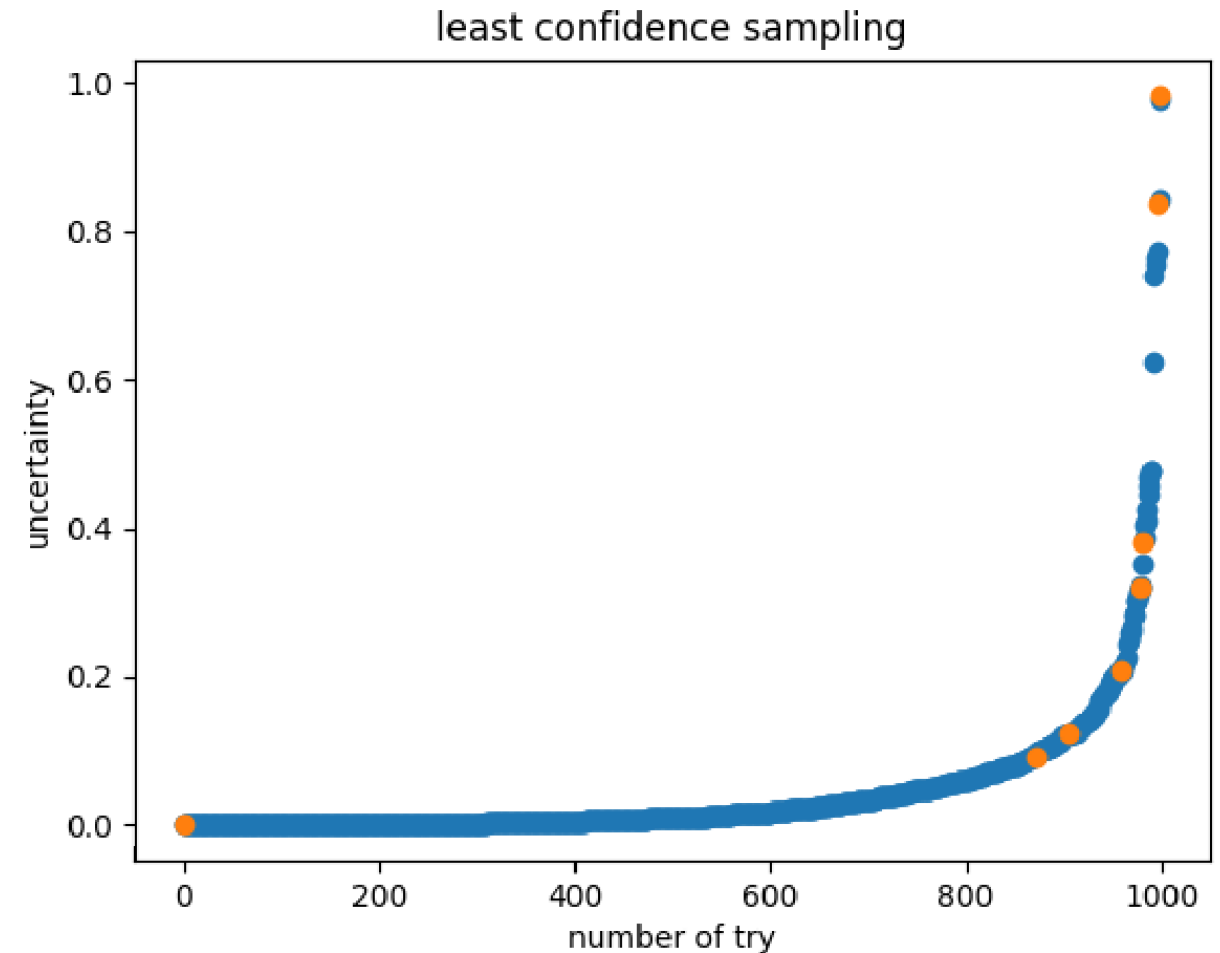
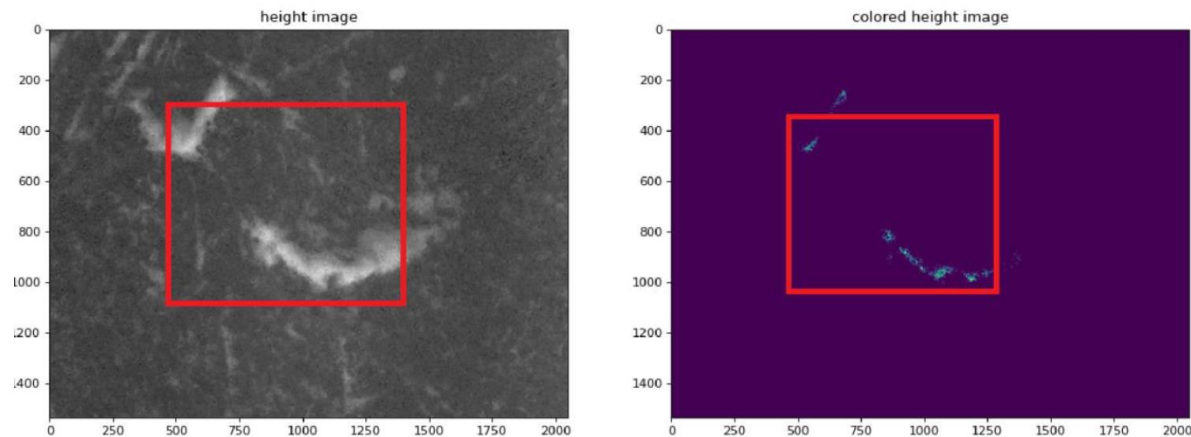


What is new?

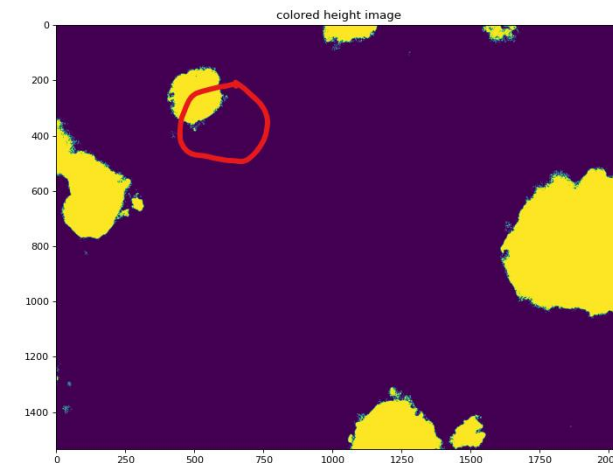
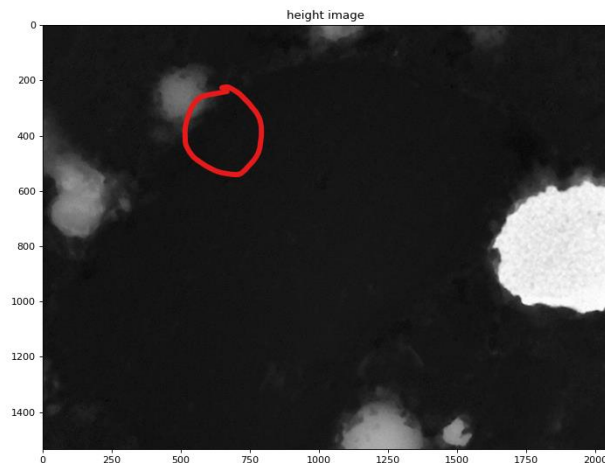
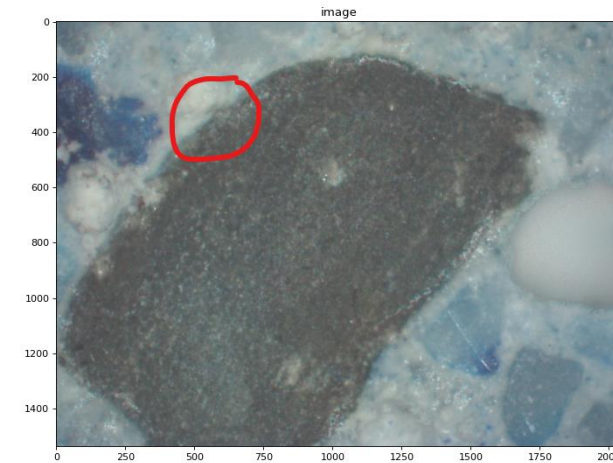
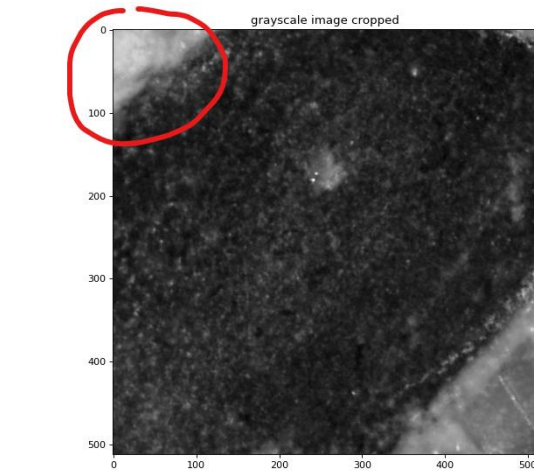


# Improvement – only aggregate classifier

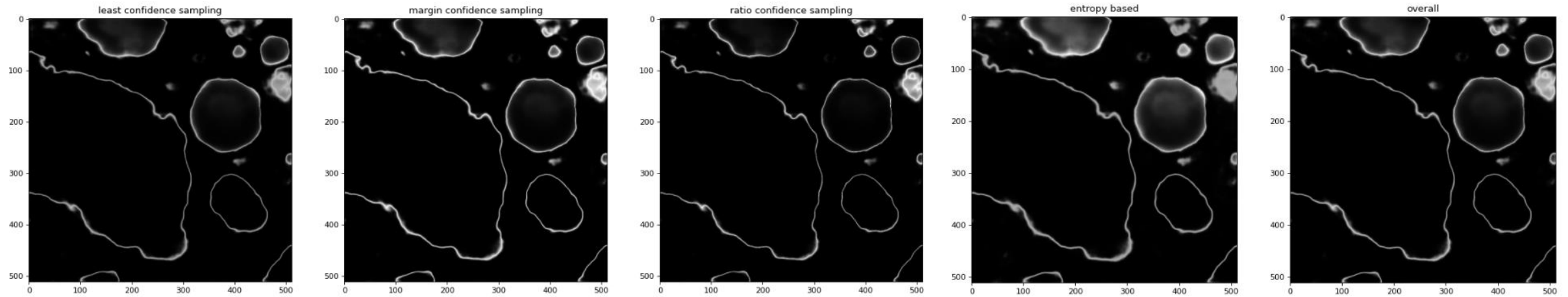
- Improved labels
- Confidence sampling analysis
- Accuracy: ~98% → >99%



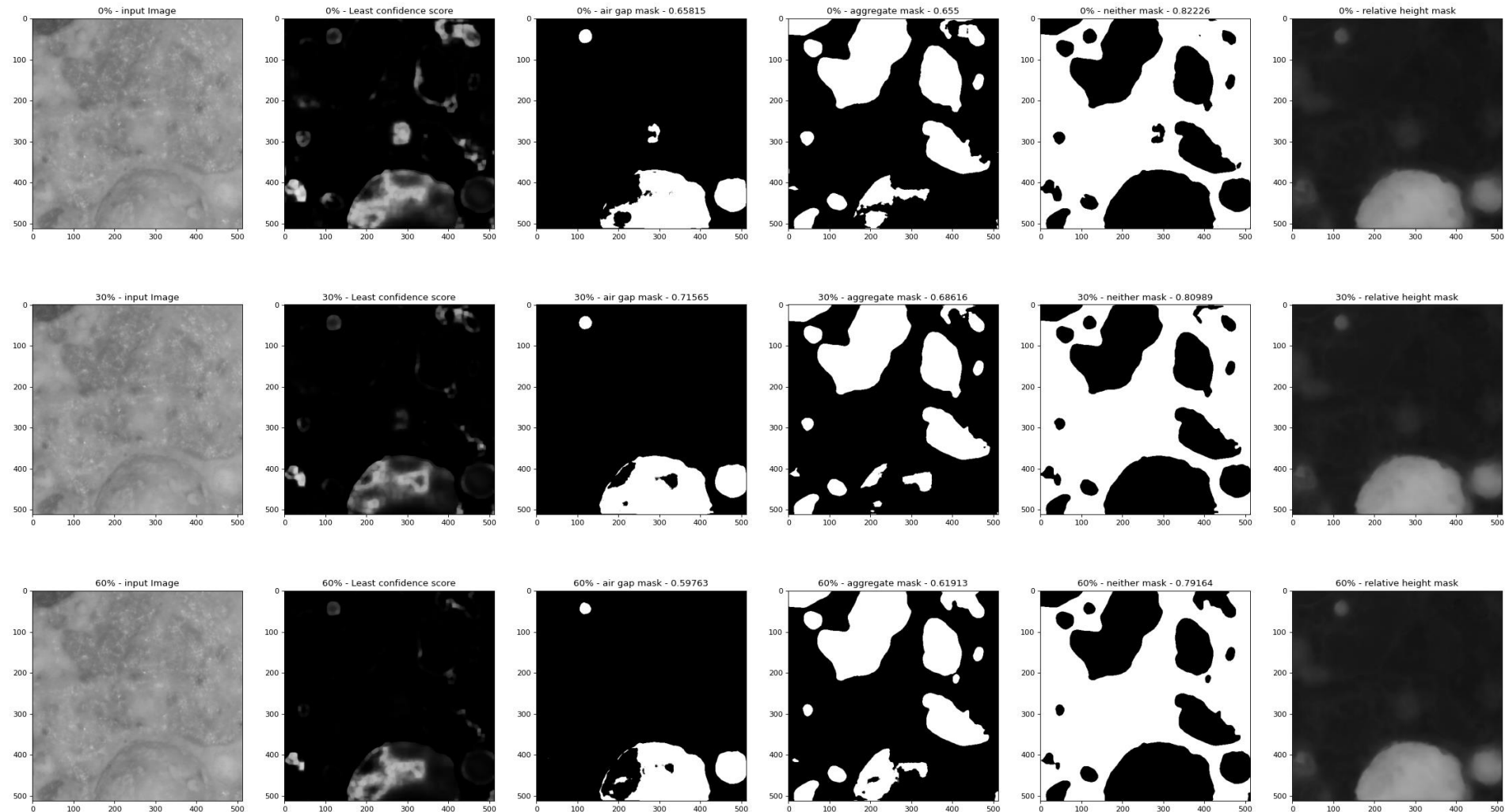
# Improvement – only aggregate classifier – Mistakes



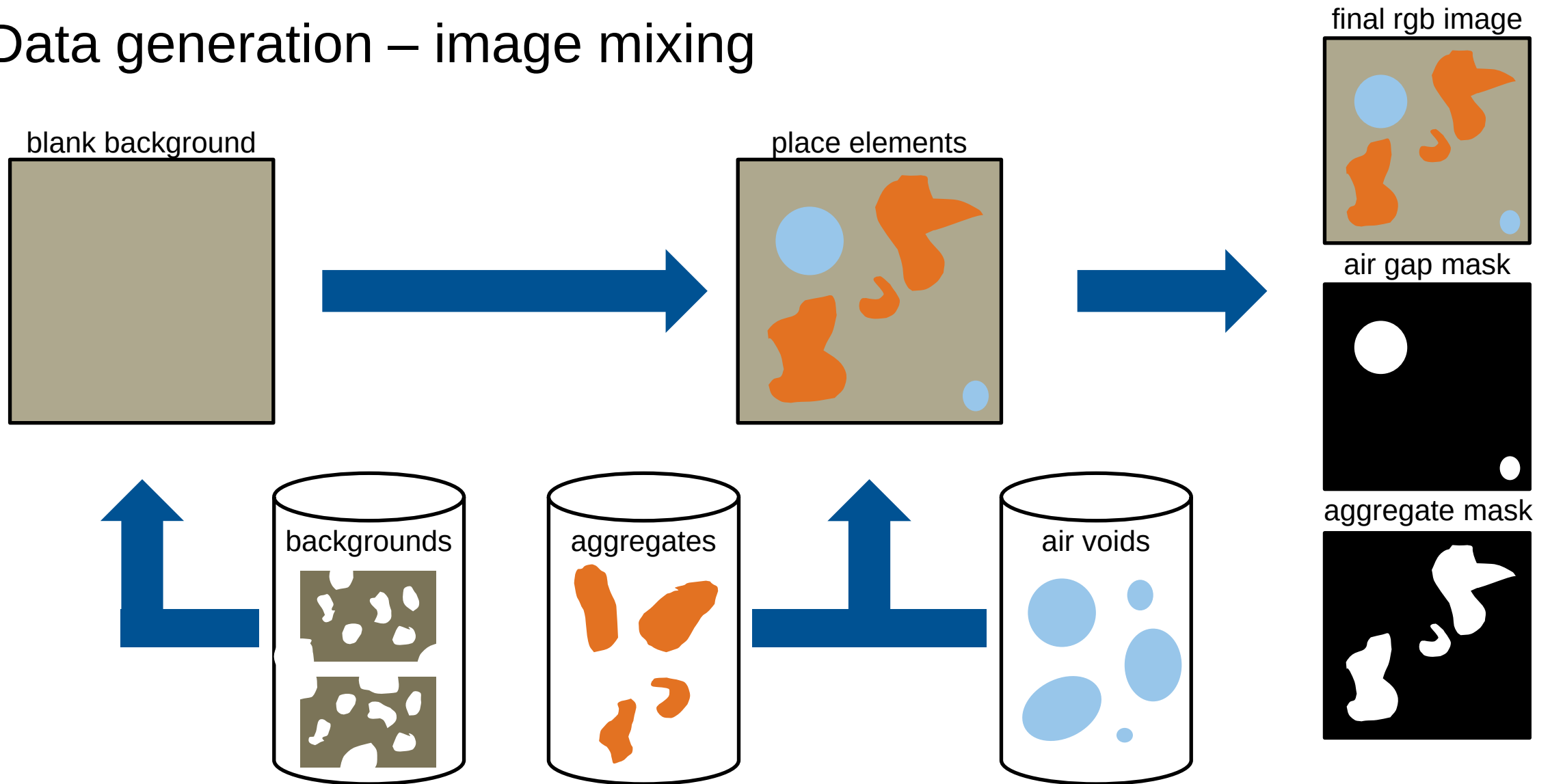
# Uncertainty sampling



# Weighted training / active learning



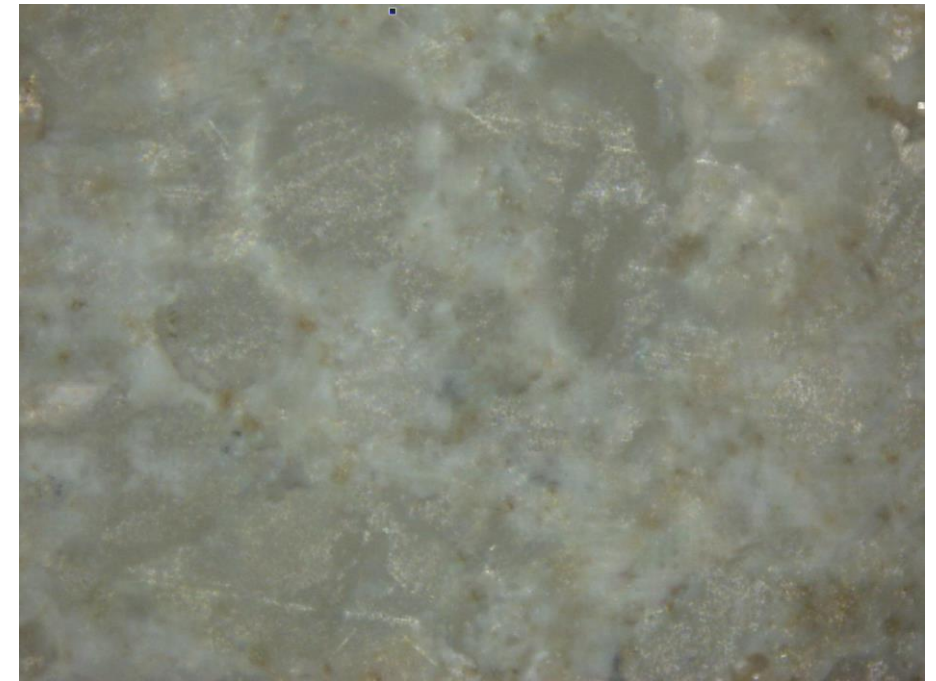
# Data generation – image mixing



# Data generation - background

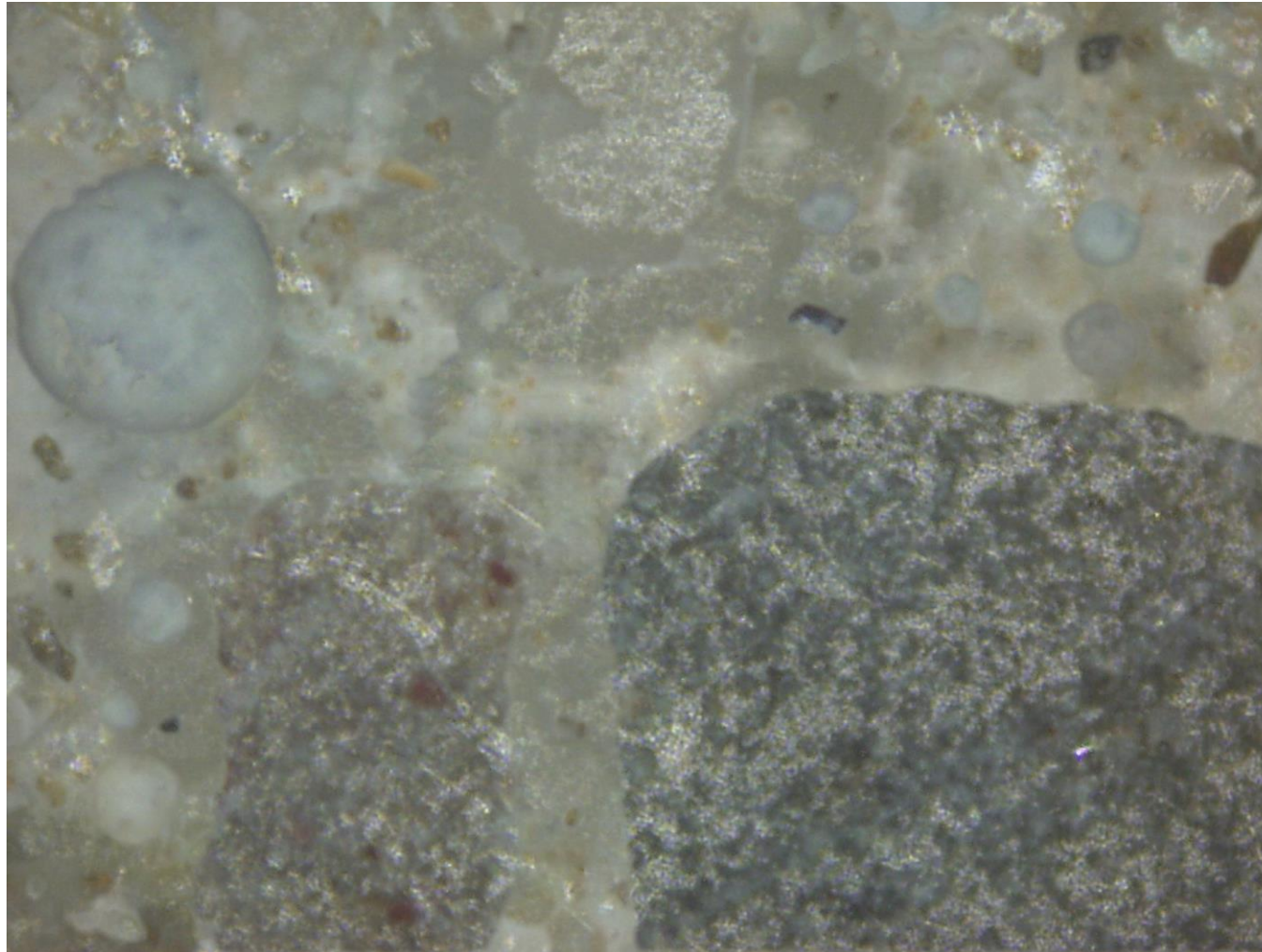


Fading | erosion  
| blemish removal





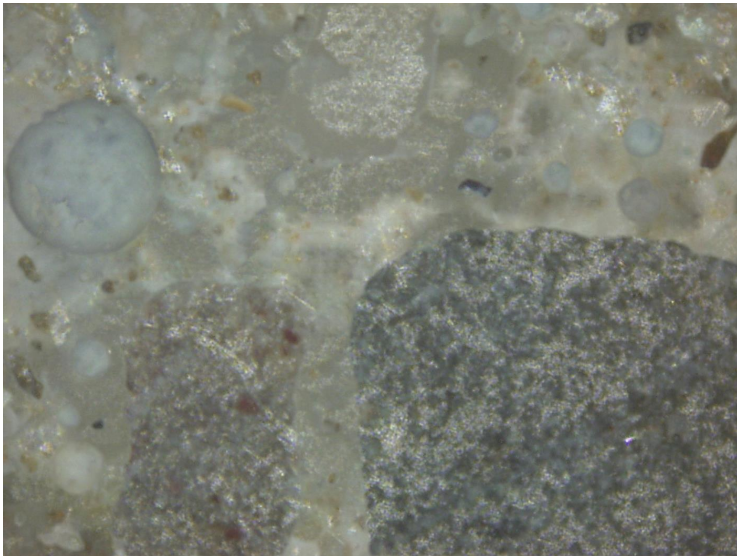
## Data generation - mixing result



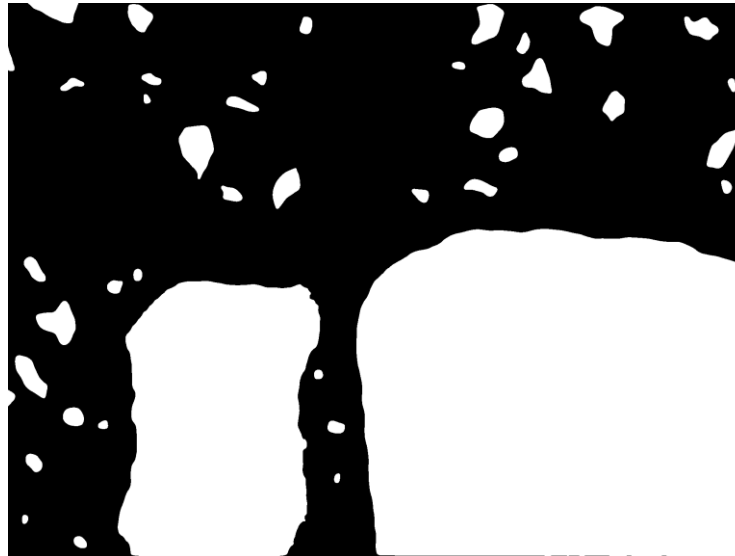


# Data generation - masks

rgb image



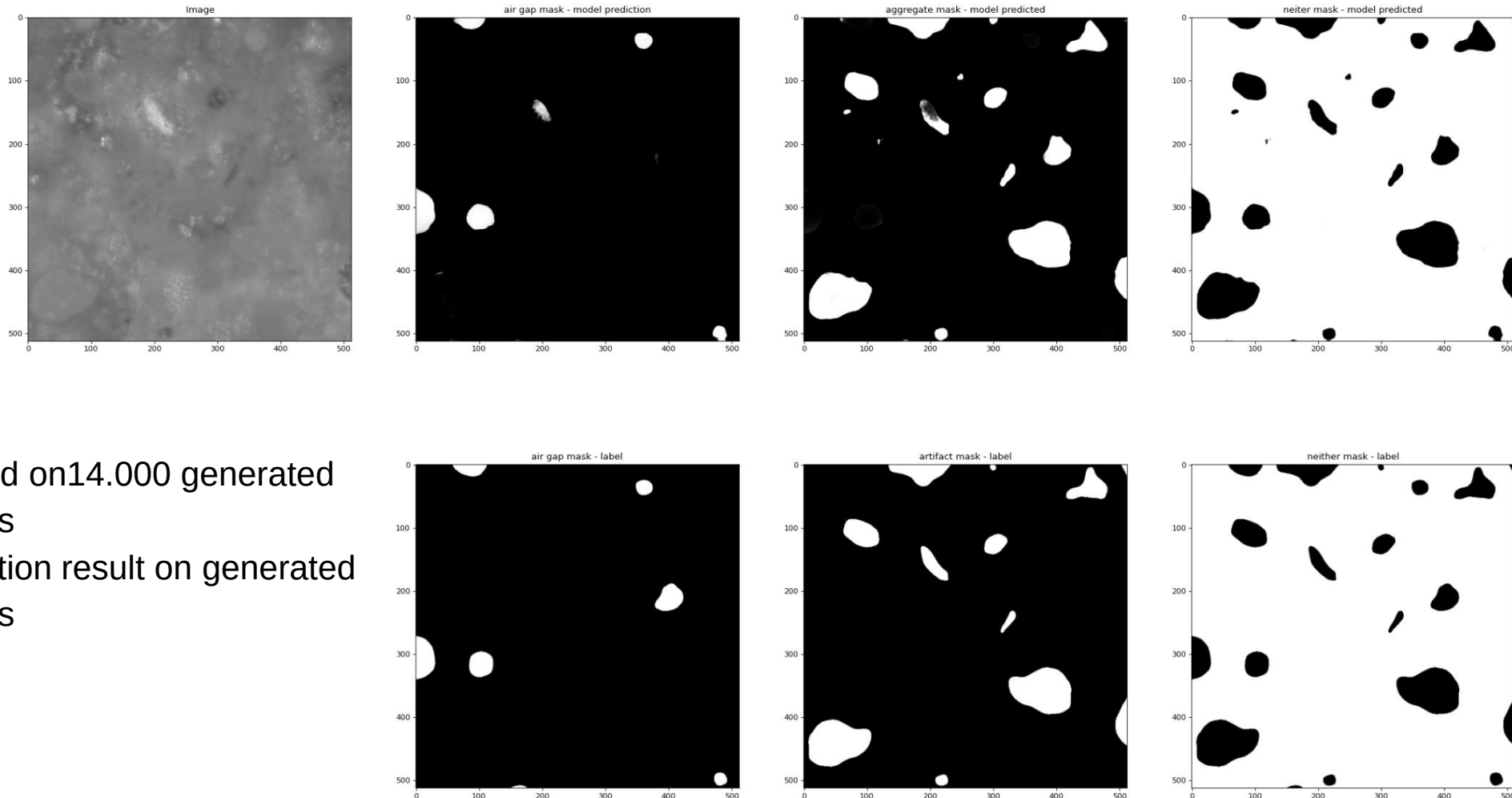
aggregate mask



air void mask

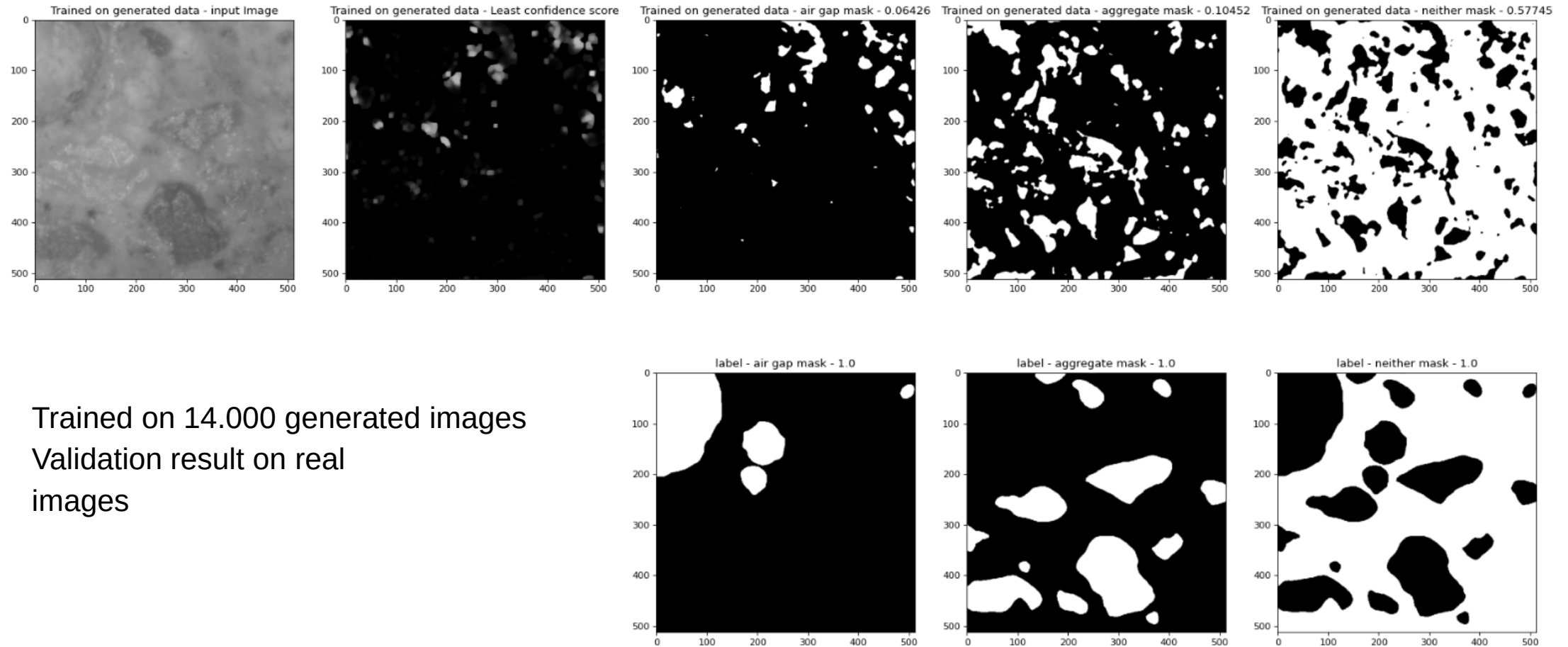


# Unet – trained on generated data



- Trained on 14.000 generated images
- Validation result on generated images

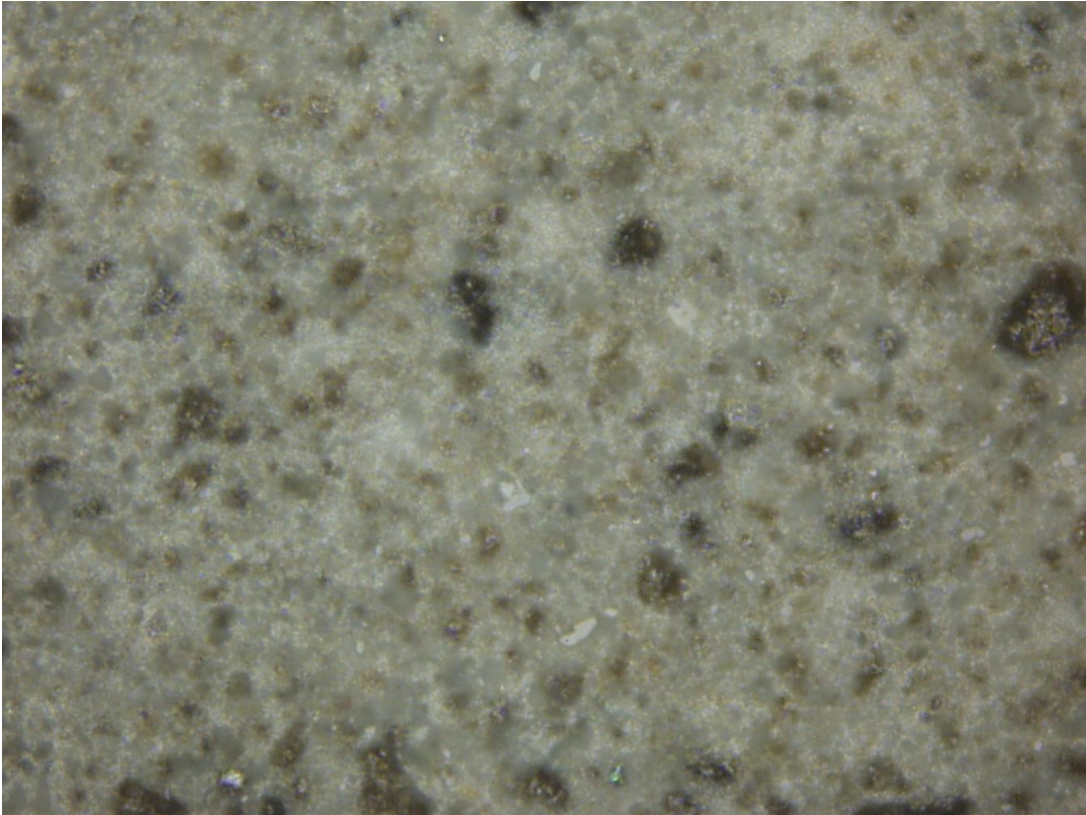
# Unet – trained on generated data



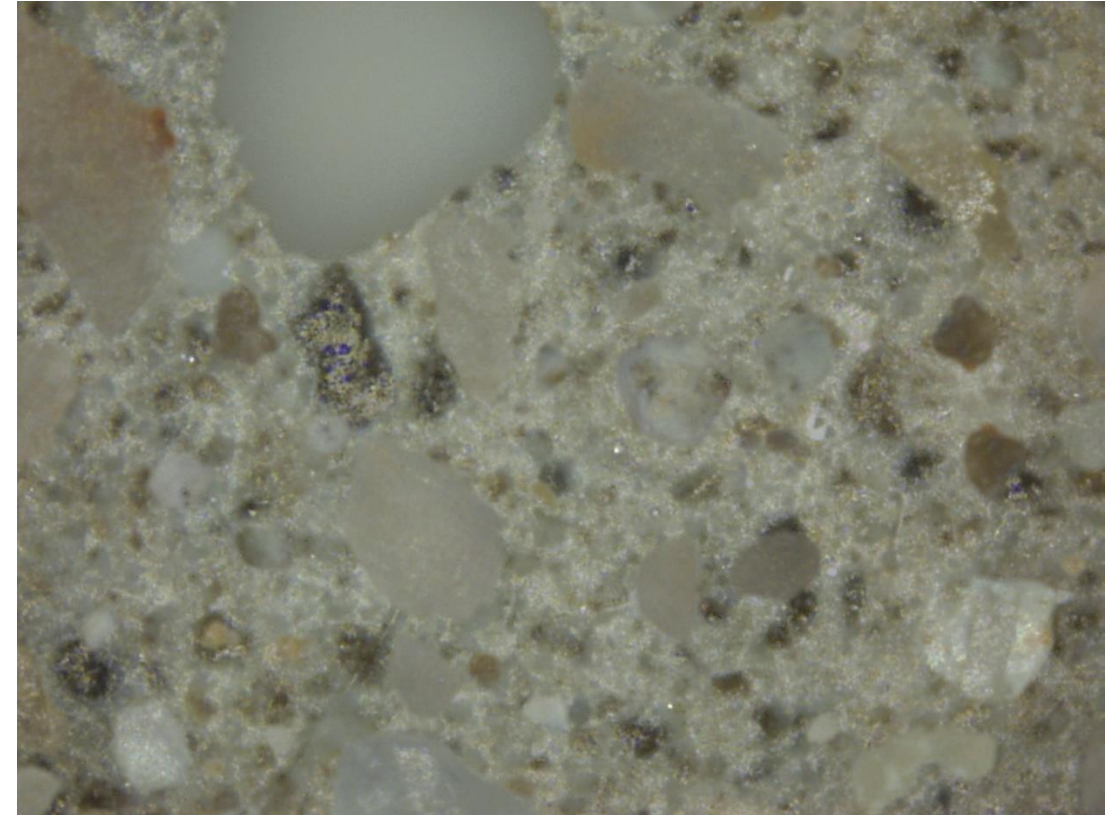
- Trained on 14.000 generated images
- Validation result on real images

# Data generation – only cement paste

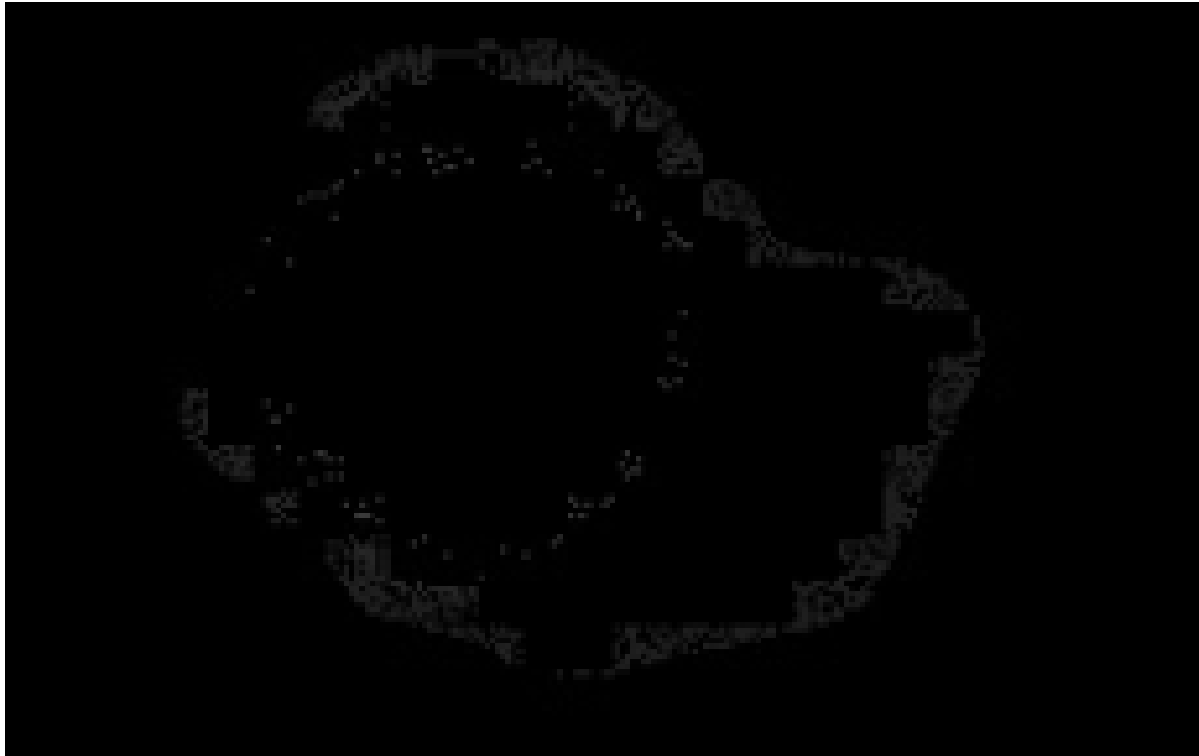
cement paste



generated image



# Improvement of manual labels

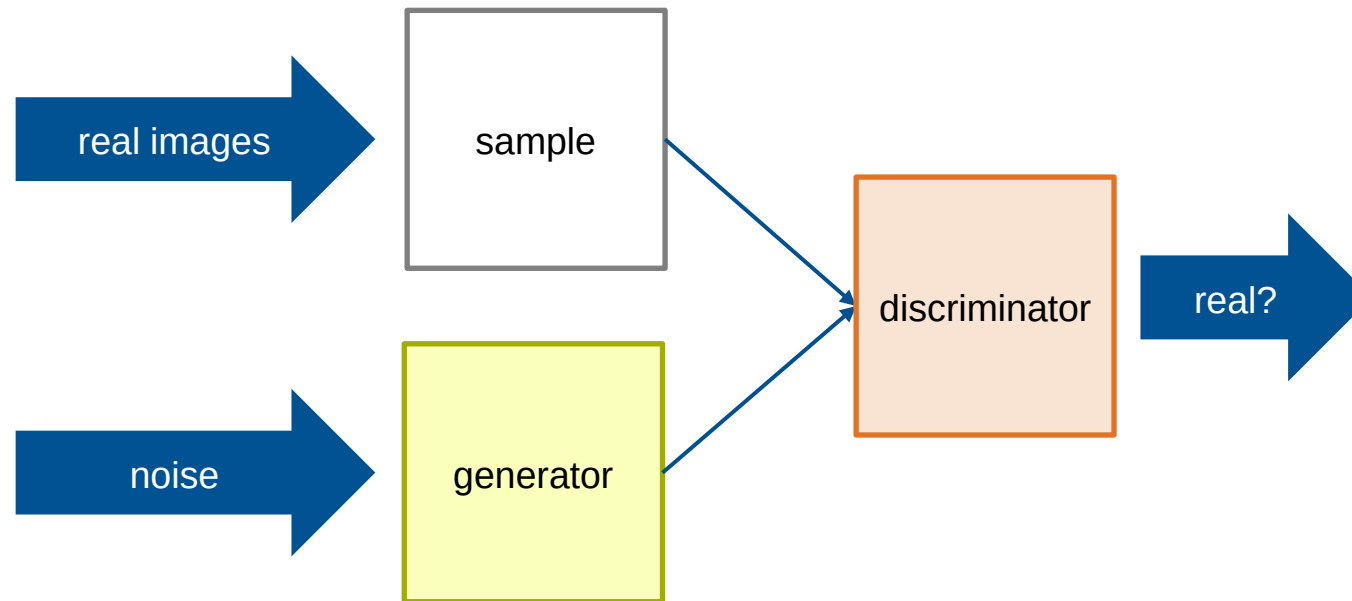


## Improved Image labels



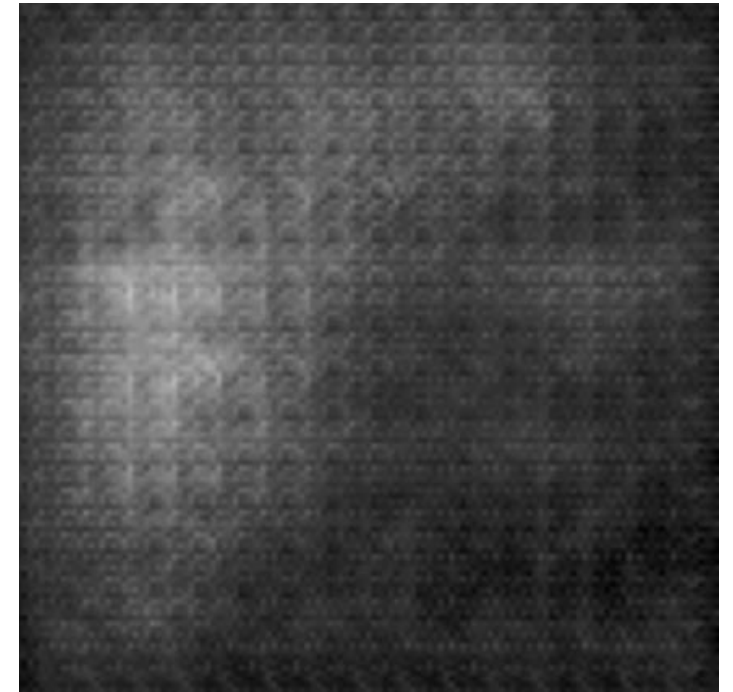
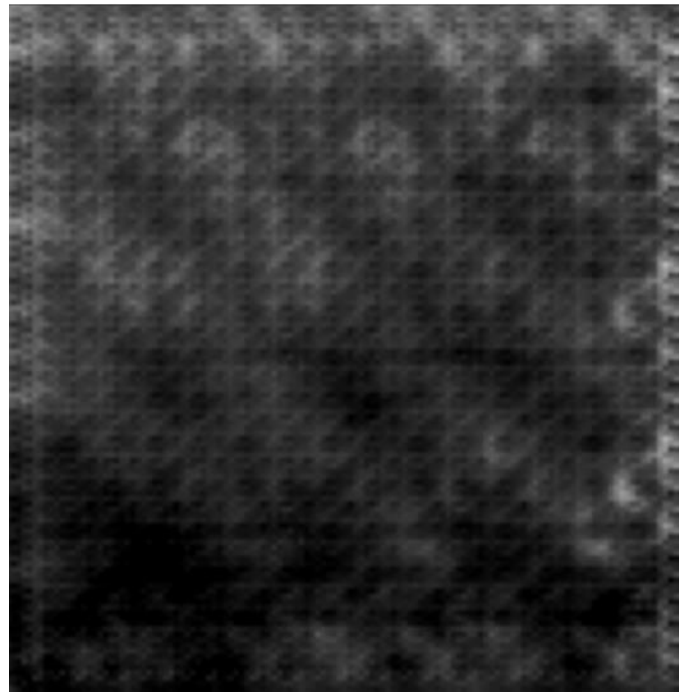
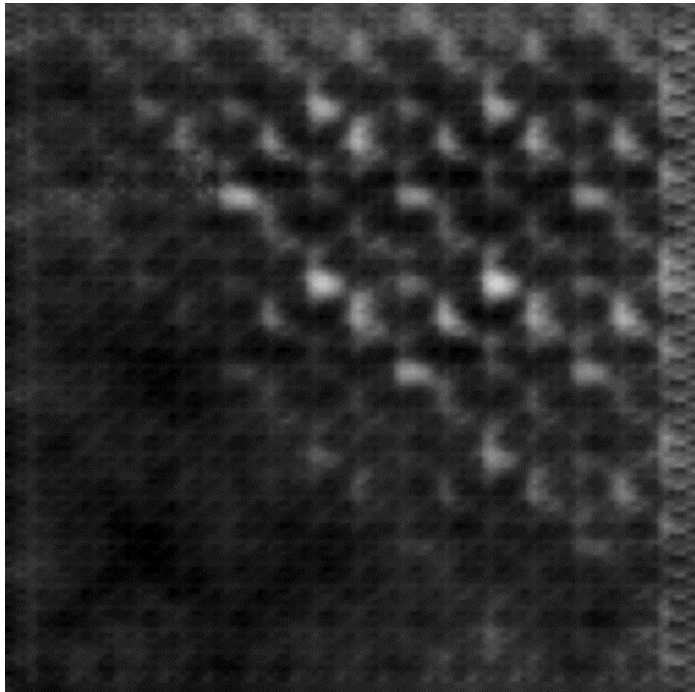


# Data generation – VAE DC GANs



# Data generation – VAE + DCGANs

- Trained on ~17k images
- Trained for 2k epochs
- Generated example images:



# Final Model – specification

## Data Augmentation:

- Color jitter: contrast, brightness, hue, saturation
- Random horizontal/vertical flip
- Random 0° | 90° | 180° | 270° rotation
- Random crop 1024px \* 1024px

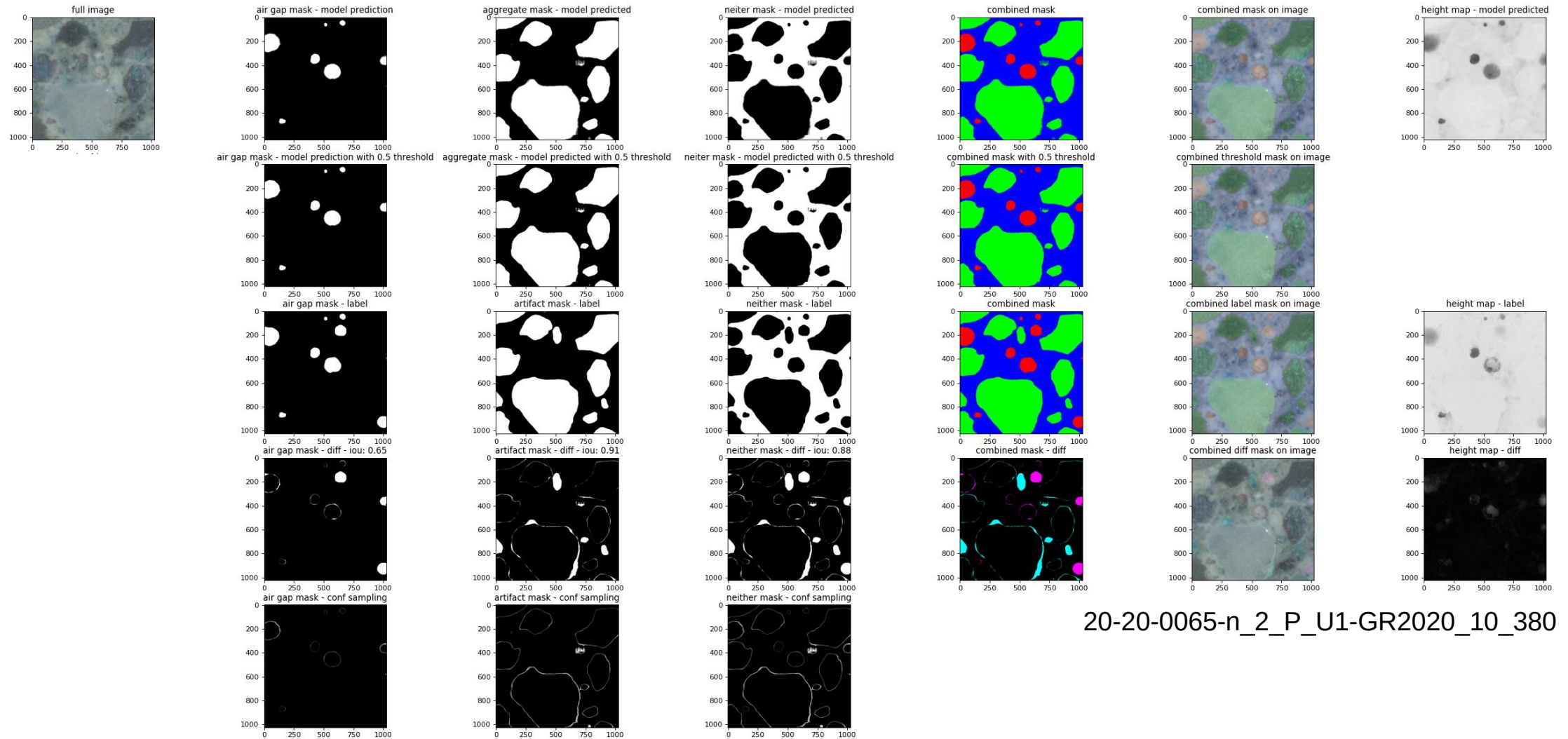
## Model:

- With padding
- Unet with 6 encode layers
- ~31M parameter
- Output: air void / aggregate / neither mask + relative height map
- No only aggregate classifier

## Dataset:

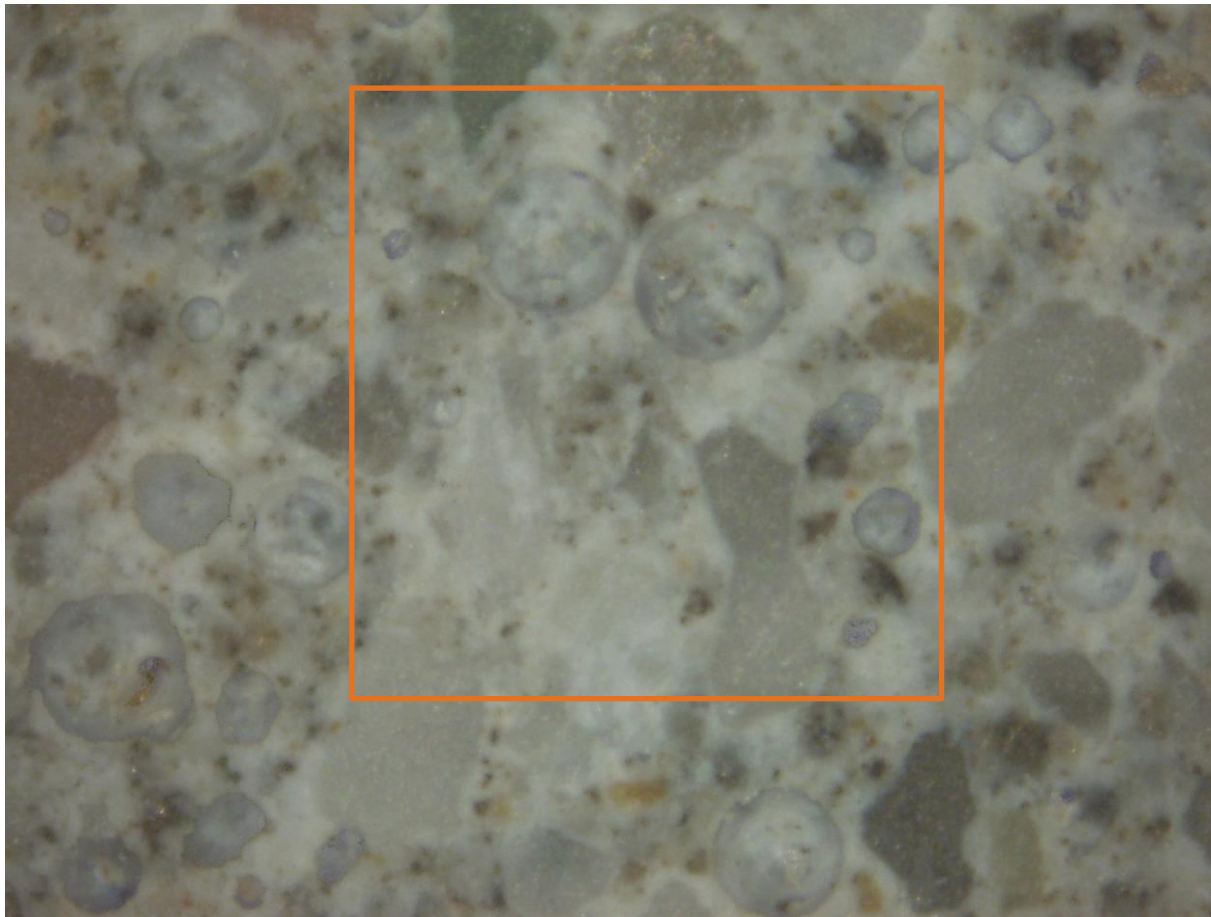
- RGB images
- Only manually corrected labels
- Diverse dataset with different concrete samples
- No tilted plain correction
- 355 training samples | 85%
- 63 validation samples | 15%

# Final Model – results

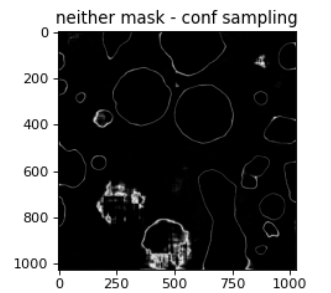
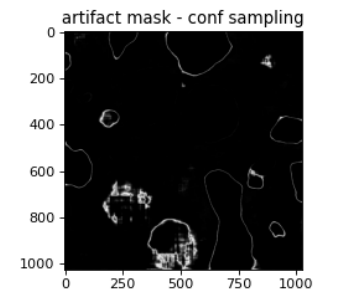
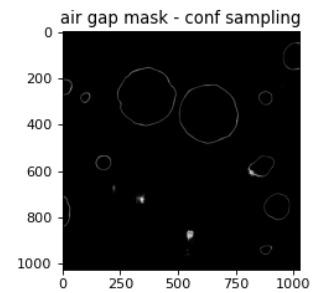
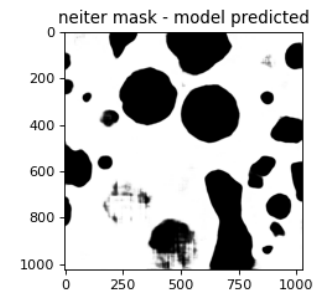
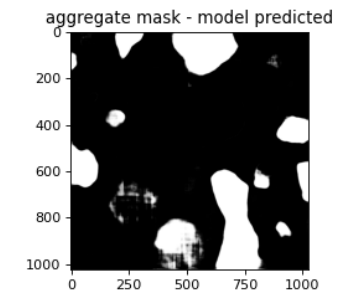
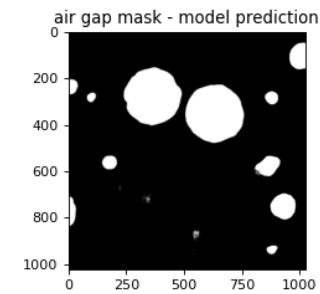


20-20-0065-n\_2\_P\_U1-GR2020\_10\_380

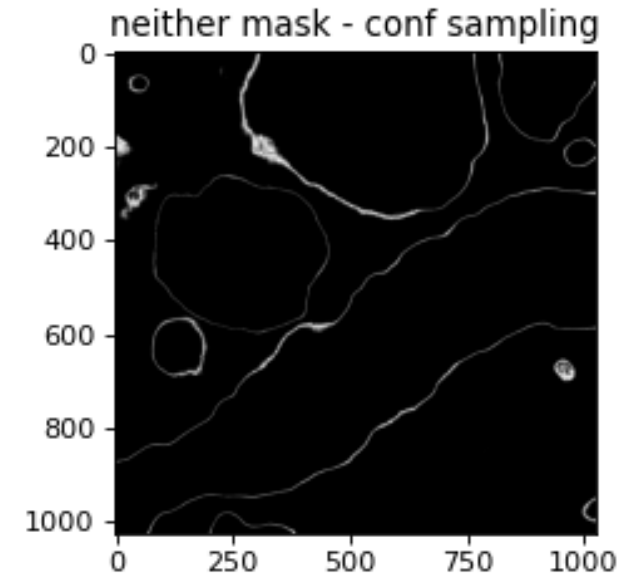
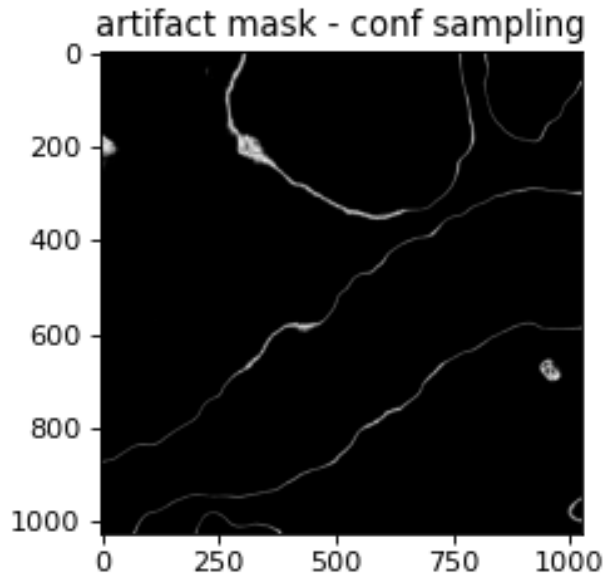
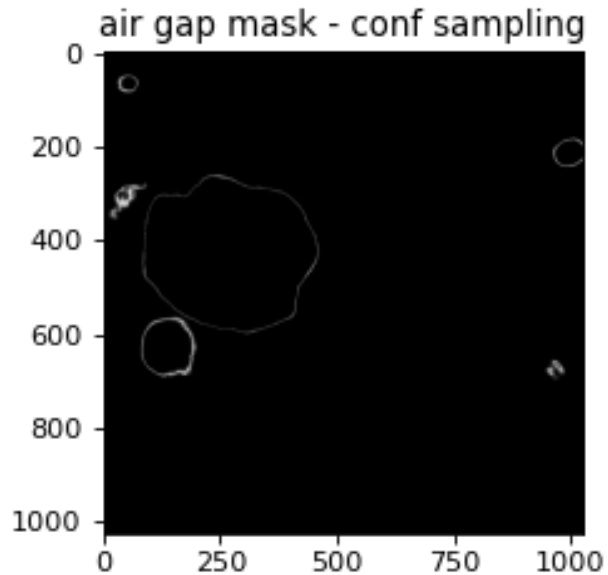
# Final Model – least confidence sampling



20-20-0065-n\_2\_P\_M2-GR2020\_10\_122



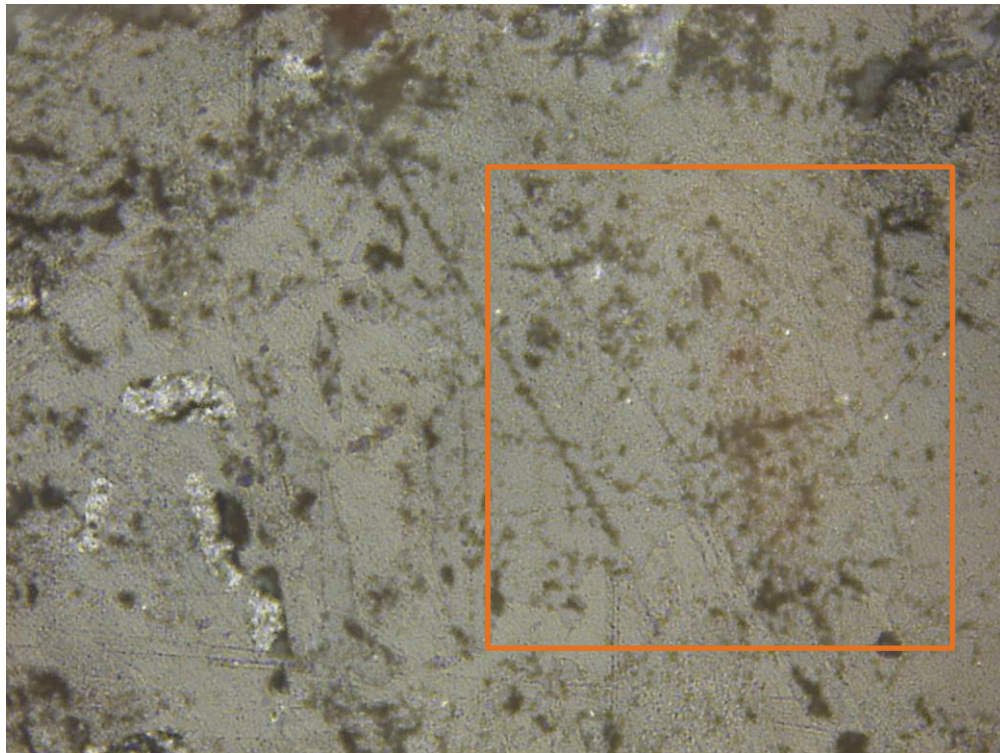
# Final Model – least confidence sampling



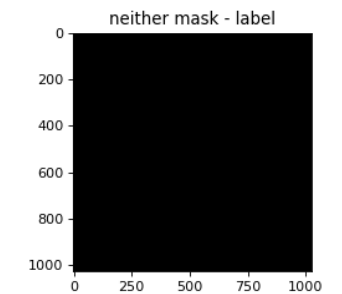
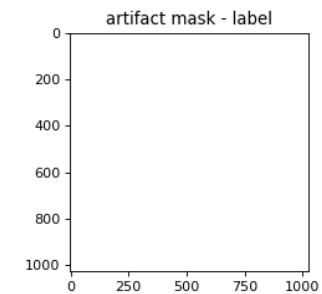
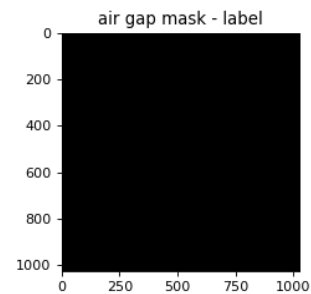
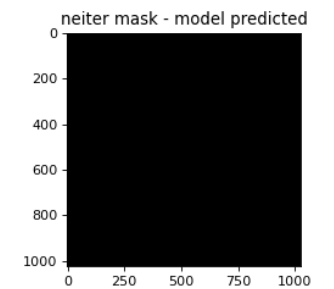
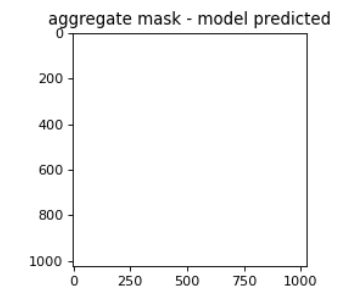
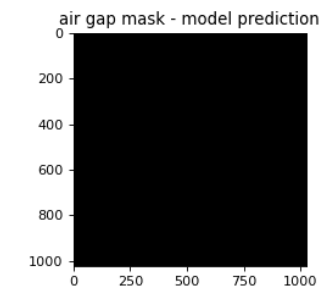
20-20-0034\_PK2\_Mitte\_26



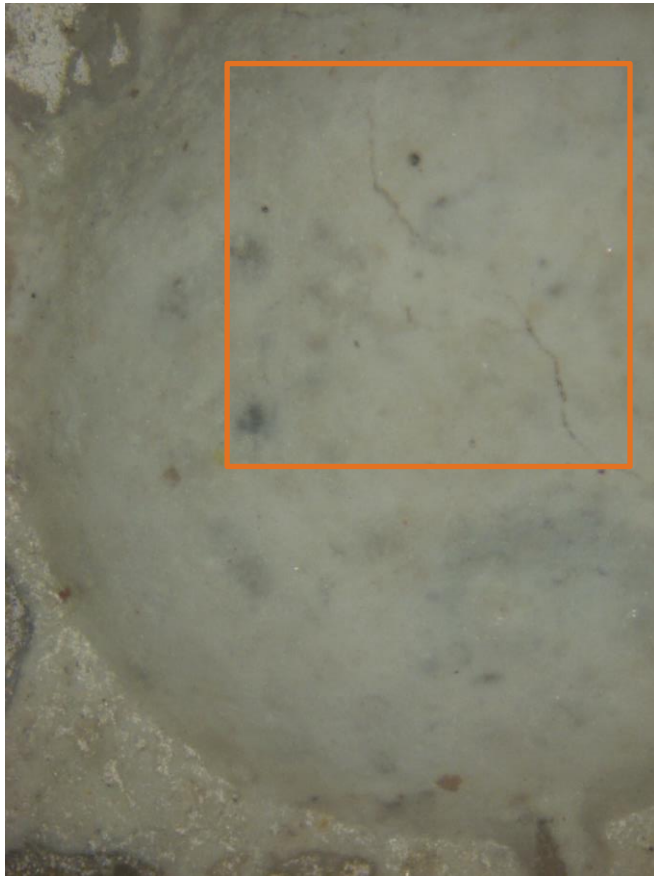
# Final Model – only aggregate



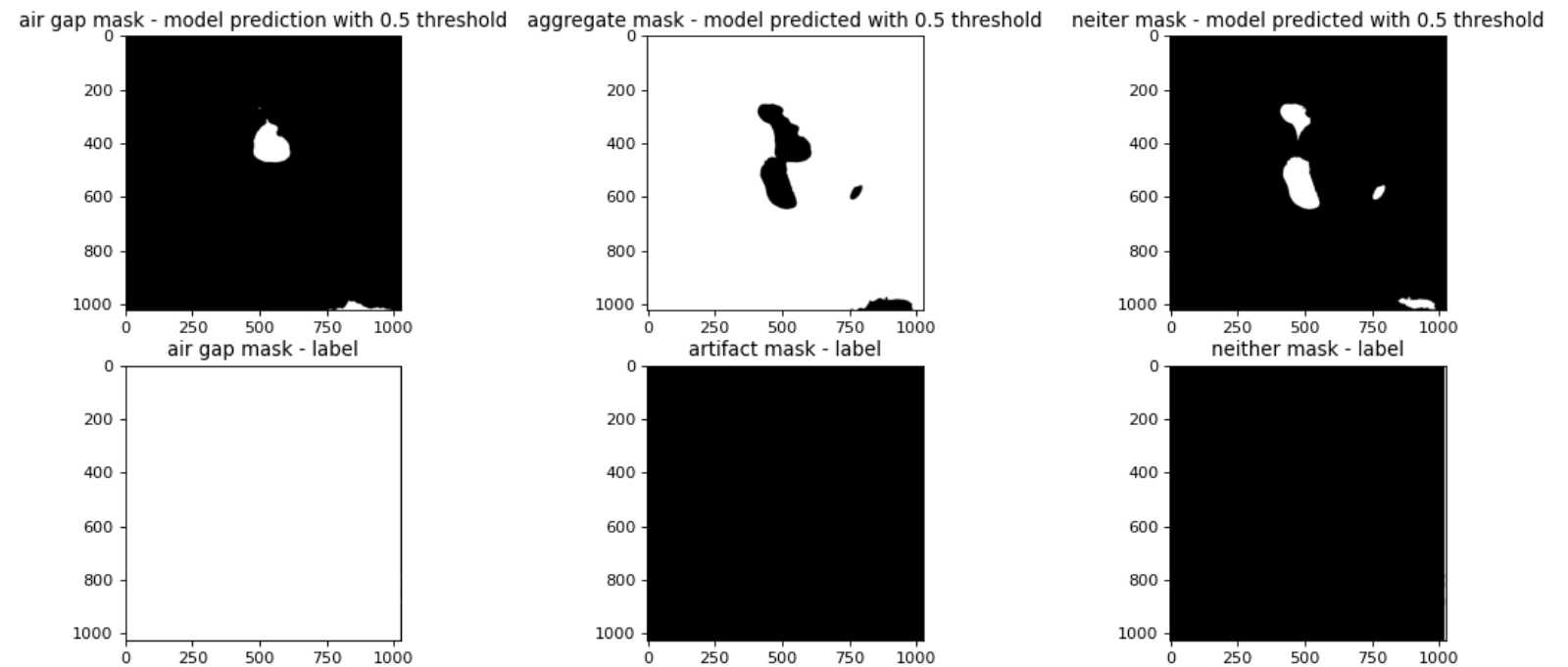
20-23-0044\_BK1.1\_P\_GR1010\_10\_80



# Final Model – large air voids



20-23-0044\_BK1.1\_P\_GR1010\_10\_26



# What now?

- Feedback loop with Viktor – What problems remain?
- Improvement number and quality of labels
- Add more edge cases to the training data
- Model/hyperparameter finetuning
- Train the final model with 100% training data
- Explore new model architectures

All Validation Results



<https://syncandshare.lrz.de/getlink/fi5Zv9rnAooUQfEJruYABt/>